CORE PURE / COMPLEX NUMBERS

De Moivre's Theorem

Worked solutions 18 questions · 184 marks

Questions

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1. Use de Moivre's theorem to find the constants A, B and C in the identity

$$\sin\theta\cos^4\theta = A\sin\theta + B\sin 3\theta + C\sin 5\theta$$ **[5]**

Solution

Let

$$z = \cos\theta + i\sin\theta$$

By de Moivre's theorem:

$$z^n = \cos n\theta + i\sin n\theta, \qquad z^{-n} = \cos n\theta - i\sin n\theta$$

so

$$z^n + z^{-n} = 2\cos n\theta, \qquad z^n - z^{-n} = 2i\sin n\theta$$

In particular, $z + z^{-1} = 2\cos\theta$ and $z - z^{-1} = 2i\sin\theta$.

Now write $\sin\theta\cos^4\theta$ in terms of z:

$$\begin{aligned}\sin\theta\cos^4\theta &= \frac{z - z^{-1}}{2i}\left(\frac{z + z^{-1}}{2}\right)^4 \\ &= \frac{1}{32i}(z - z^{-1})(z + z^{-1})^4\end{aligned}$$

Expand $(z + z^{-1})^4$:

$$(z + z^{-1})^4 = z^4 + 4z^2 + 6 + 4z^{-2} + z^{-4}$$

So

$$\begin{aligned}\sin\theta\cos^4\theta &= \frac{1}{32i}(z - z^{-1})(z^4 + 4z^2 + 6 + 4z^{-2} + z^{-4}) \\ &= \frac{1}{32i}\big(z^5 + 4z^3 + 6z + 4z^{-1} + z^{-3} \\ &\qquad - z^3 - 4z - 6z^{-1} - 4z^{-3} - z^{-5}\big) \\ &= \frac{1}{32i}\big(z^5 + 3z^3 + 2z - 2z^{-1} - 3z^{-3} - z^{-5}\big) \\ &= \frac{1}{32i}\Big[(z^5 - z^{-5}) + 3(z^3 - z^{-3}) + 2(z - z^{-1})\Big]\end{aligned}$$

Using $z^n - z^{-n} = 2i\sin n\theta$:

$$\begin{aligned}\sin\theta\cos^4\theta &= \frac{1}{16}\sin 5\theta + \frac{3}{16}\sin 3\theta + \frac{2}{16}\sin\theta \\ &= \frac{1}{8}\sin\theta + \frac{3}{16}\sin 3\theta + \frac{1}{16}\sin 5\theta\end{aligned}$$

Comparing with

$$\sin\theta\cos^4\theta = A\sin\theta + B\sin 3\theta + C\sin 5\theta$$

gives

$$A = \frac{1}{8}, \qquad B = \frac{3}{16}, \qquad C = \frac{1}{16}$$

2. A complex number z has modulus 1 and argument θ.

(a) Show that

$$z^n + \frac{1}{z^n} = 2\cos n\theta \quad \text{and} \quad z^n - \frac{1}{z^n} = 2i\sin n\theta, \qquad n \in \mathbb{Z}^+$$ **[3]**

(b) Hence, by expanding $\left(z + \frac{1}{z}\right)^3$ and $\left(z - \frac{1}{z}\right)^3$, show that

$$\tan 3\theta = \frac{3\tan\theta - \tan^3\theta}{1 - 3\tan^2\theta}$$ **[4]**

Solution

(a) Since $|z| = 1$ and $\arg z = \theta$, we have

$$z = \cos\theta + i\sin\theta$$

So, by de Moivre's theorem:

$$z^n = \cos n\theta + i\sin n\theta$$

and

$$\frac{1}{z^n} = z^{-n} = \cos(-n\theta) + i\sin(-n\theta) = \cos n\theta - i\sin n\theta$$

Now add:

$$z^n + \frac{1}{z^n} = (\cos n\theta + i\sin n\theta) + (\cos n\theta - i\sin n\theta) = 2\cos n\theta$$

Now subtract:

$$z^n - \frac{1}{z^n} = (\cos n\theta + i\sin n\theta) - (\cos n\theta - i\sin n\theta) = 2i\sin n\theta$$

Therefore

$$z^n + \frac{1}{z^n} = 2\cos n\theta \quad \text{and} \quad z^n - \frac{1}{z^n} = 2i\sin n\theta$$

(b) From part (a), with $n = 1$:

$$z + \frac{1}{z} = 2\cos\theta$$

Cubing both sides gives

$$\left(z + \frac{1}{z}\right)^3 = (2\cos\theta)^3 = 8\cos^3\theta$$

Expand the left-hand side:

$$\begin{aligned}\left(z + \frac{1}{z}\right)^3 &= z^3 + 3z + \frac{3}{z} + \frac{1}{z^3} \\ &= \left(z^3 + \frac{1}{z^3}\right) + 3\left(z + \frac{1}{z}\right)\end{aligned}$$

Using part (a) again:

$$8\cos^3\theta = 2\cos 3\theta + 6\cos\theta$$

So

$$\cos 3\theta = 4\cos^3\theta - 3\cos\theta$$

Now use part (a) with

$$z - \frac{1}{z} = 2i\sin\theta$$

Cubing both sides:

$$\left(z - \frac{1}{z}\right)^3 = (2i\sin\theta)^3 = -8i\sin^3\theta$$

Expand:

$$\begin{aligned}\left(z - \frac{1}{z}\right)^3 &= z^3 - 3z + \frac{3}{z} - \frac{1}{z^3} \\ &= \left(z^3 - \frac{1}{z^3}\right) - 3\left(z - \frac{1}{z}\right)\end{aligned}$$

Using part (a):

$$-8i\sin^3\theta = 2i\sin 3\theta - 6i\sin\theta$$

Hence

$$\sin 3\theta = 3\sin\theta - 4\sin^3\theta$$

Let $t = \tan\theta$. Then

$$\begin{aligned}\tan 3\theta &= \frac{\sin 3\theta}{\cos 3\theta} \\ &= \frac{\dfrac{\sin 3\theta}{\cos^3\theta}}{\dfrac{\cos 3\theta}{\cos^3\theta}} \\ &= \frac{\dfrac{3\sin\theta - 4\sin^3\theta}{\cos^3\theta}}{\dfrac{4\cos^3\theta - 3\cos\theta}{\cos^3\theta}} \\ &= \frac{3\tan\theta\sec^2\theta - 4\tan^3\theta}{4 - 3\sec^2\theta} \\ &= \frac{3t(1+t^2) - 4t^3}{4 - 3(1+t^2)} \\ &= \frac{3t - t^3}{1 - 3t^2}\end{aligned}$$

Therefore

$$\tan 3\theta = \frac{3\tan\theta - \tan^3\theta}{1 - 3\tan^2\theta}$$

3. Let $t = \tan\theta$. By writing

$$\cos\theta + i\sin\theta = \cos\theta\,(1 + it)$$

and expanding $(1 + it)^6$, use de Moivre's theorem to show that

$$\tan 6\theta = \frac{6\tan\theta - 20\tan^3\theta + 6\tan^5\theta}{1 - 15\tan^2\theta + 15\tan^4\theta - \tan^6\theta} \qquad \textbf{[7]}$$

Solution

Let

$$z = \cos\theta + i\sin\theta$$

and let

$$t = \tan\theta = \frac{\sin\theta}{\cos\theta}$$

Then $\sin\theta = t\cos\theta$, so

$$z = \cos\theta + i\sin\theta = \cos\theta + i(t\cos\theta) = \cos\theta(1 + it)$$

Now apply de Moivre's theorem:

$$z^6 = \cos 6\theta + i\sin 6\theta$$

But also, from $z = \cos\theta(1 + it)$:

$$z^6 = \cos^6\theta\,(1 + it)^6$$

So we expand $(1 + it)^6$ using the binomial theorem:

$$(1 + it)^6 = 1 + 6(it) + 15(it)^2 + 20(it)^3 + 15(it)^4 + 6(it)^5 + (it)^6$$

Now simplify the powers of i:

$$i^2 = -1, \qquad i^3 = -i, \qquad i^4 = 1, \qquad i^5 = i, \qquad i^6 = -1$$

Hence

$$\begin{aligned}(1 + it)^6 &= 1 + 6it + 15(-1)t^2 + 20(-i)t^3 + 15(1)t^4 + 6(i)t^5 + (-1)t^6\\ &= 1 + 6it - 15t^2 - 20it^3 + 15t^4 + 6it^5 - t^6\\ &= (1 - 15t^2 + 15t^4 - t^6) + i(6t - 20t^3 + 6t^5)\end{aligned}$$

Therefore

$$\cos 6\theta + i\sin 6\theta = \cos^6\theta\Big[(1 - 15t^2 + 15t^4 - t^6) + i(6t - 20t^3 + 6t^5)\Big]$$

Equating real and imaginary parts gives

$$\cos 6\theta = \cos^6\theta(1 - 15t^2 + 15t^4 - t^6)$$

and

$$\sin 6\theta = \cos^6\theta(6t - 20t^3 + 6t^5)$$

So

$$\begin{aligned}\tan 6\theta &= \frac{\sin 6\theta}{\cos 6\theta} \\ &= \frac{\cos^6\theta(6t - 20t^3 + 6t^5)}{\cos^6\theta(1 - 15t^2 + 15t^4 - t^6)} \\ &= \frac{6t - 20t^3 + 6t^5}{1 - 15t^2 + 15t^4 - t^6}\end{aligned}$$

Since $t = \tan\theta$, this becomes

$$\tan 6\theta = \frac{6\tan\theta - 20\tan^3\theta + 6\tan^5\theta}{1 - 15\tan^2\theta + 15\tan^4\theta - \tan^6\theta}$$

as required.

4. **(a)** Use de Moivre's theorem to show that

$$\sin 6x = \sin 2x \left(a\cos^4 x + b\cos^2 x + c\right)$$

where a, b and c are integers to be determined. **[5]**

(b) Hence solve, for $0 < \theta < \frac{\pi}{2}$,

$$\sin 6\theta = \sin 2\theta \cos 2\theta$$

giving your answers to 3 decimal places. **[4]**

Solution

(a) By de Moivre's theorem:

$$(\cos x + i\sin x)^6 = \cos 6x + i\sin 6x$$

So

$$\begin{aligned}(\cos x + i\sin x)^6 &= \cos^6 x + 6i\cos^5 x\sin x + 15\cos^4 x(i\sin x)^2 + 20\cos^3 x(i\sin x)^3 \\ &\quad + 15\cos^2 x(i\sin x)^4 + 6\cos x(i\sin x)^5 + (i\sin x)^6 \\ &= \cos^6 x + 6i\cos^5 x\sin x - 15\cos^4 x\sin^2 x - 20i\cos^3 x\sin^3 x \\ &\quad + 15\cos^2 x\sin^4 x + 6i\cos x\sin^5 x - \sin^6 x\end{aligned}$$

Equating imaginary parts gives

$$\begin{aligned}\sin 6x &= 6\cos^5 x\sin x - 20\cos^3 x\sin^3 x + 6\cos x\sin^5 x \\ &= 2\sin x\cos x\left(3\cos^4 x - 10\cos^2 x\sin^2 x + 3\sin^4 x\right)\end{aligned}$$

Now simplify the bracket:

$$\begin{aligned}3\cos^4 x - 10\cos^2 x\sin^2 x + 3\sin^4 x &= 3(\cos^4 x + \sin^4 x) - 10\cos^2 x\sin^2 x \\ &= 3\left((\cos^2 x + \sin^2 x)^2 - 2\cos^2 x\sin^2 x\right) - 10\cos^2 x\sin^2 x \\ &= 3(1 - 2\cos^2 x\sin^2 x) - 10\cos^2 x\sin^2 x \\ &= 3 - 16\cos^2 x\sin^2 x \\ &= 3 - 16\cos^2 x(1 - \cos^2 x) \\ &= 16\cos^4 x - 16\cos^2 x + 3\end{aligned}$$

Also, $\sin 2x = 2\sin x\cos x$, so

$$\sin 6x = \sin 2x\left(16\cos^4 x - 16\cos^2 x + 3\right)$$

Hence

$$a = 16, \qquad b = -16, \qquad c = 3$$

(b) Using part (a):

$$\sin 6\theta = \sin 2\theta\left(16\cos^4\theta - 16\cos^2\theta + 3\right)$$

So the equation

$$\sin 6\theta = \sin 2\theta \cos 2\theta$$

becomes

$$\sin 2\theta\big(16\cos^4\theta - 16\cos^2\theta + 3\big) = \sin 2\theta \cos 2\theta$$

Since $0 < \theta < \frac{\pi}{2}$, we have $0 < 2\theta < \pi$, so $\sin 2\theta \neq 0$.

Therefore we can divide by $\sin 2\theta$:

$$16\cos^4\theta - 16\cos^2\theta + 3 = \cos 2\theta$$

Let

$$u = \cos^2\theta$$

Then $\cos 2\theta = 2u - 1$, so

$$16u^2 - 16u + 3 = 2u - 1$$

$$16u^2 - 18u + 4 = 0$$

so

$$u = \frac{9 \pm \sqrt{17}}{16}$$

Hence

$$\begin{aligned}\cos 2\theta &= 2u - 1 \\ &= 2\left(\frac{9 \pm \sqrt{17}}{16}\right) - 1 \\ &= \frac{1 \pm \sqrt{17}}{8}\end{aligned}$$

So

$$2\theta = \arccos\left(\frac{1 \pm \sqrt{17}}{8}\right)$$

and therefore

$$\theta = \frac{1}{2}\arccos\left(\frac{1 \pm \sqrt{17}}{8}\right)$$

This gives

$$\theta \approx 0.438,\ 0.986 \qquad (3 \text{ d.p.})$$

5. **(a)** By considering the binomial expansion of $(z - z^{-1})^6$, where $z = \cos\theta + i\sin\theta$, use de Moivre's theorem to show that

$$\sin^6\theta = \frac{1}{32}\big(10 - 15\cos 2\theta + 6\cos 4\theta - \cos 6\theta\big) \qquad \textbf{[5]}$$

(b) Hence, using the substitution $x = \tan\theta$, find the exact value of

$$\int_0^1 \frac{x^6}{(1+x^2)^4}\,\mathrm{d}x \qquad \textbf{[6]}$$

Solution

(a) By de Moivre's theorem:

$$z^n = \cos n\theta + i\sin n\theta, \qquad z^{-n} = \cos n\theta - i\sin n\theta$$

so

$$z^n + z^{-n} = 2\cos n\theta, \qquad z^n - z^{-n} = 2i\sin n\theta$$

In particular, $z - z^{-1} = 2i\sin\theta$, so

$$(z - z^{-1})^6 = (2i\sin\theta)^6 = 64i^6\sin^6\theta = -64\sin^6\theta$$

Now expand $(z - z^{-1})^6$ by the binomial theorem:

$$\begin{aligned}(z - z^{-1})^6 &= z^6 - 6z^5(z^{-1}) + 15z^4(z^{-1})^2 - 20z^3(z^{-1})^3 \\ &\quad + 15z^2(z^{-1})^4 - 6z(z^{-1})^5 + (z^{-1})^6 \\ &= z^6 - 6z^4 + 15z^2 - 20 + 15z^{-2} - 6z^{-4} + z^{-6}\end{aligned}$$

Therefore

$$\begin{aligned}(z - z^{-1})^6 &= (z^6 + z^{-6}) - 6(z^4 + z^{-4}) + 15(z^2 + z^{-2}) - 20 \\ &= 2\cos 6\theta - 12\cos 4\theta + 30\cos 2\theta - 20\end{aligned}$$

Equating this with the earlier result gives

$$-64\sin^6\theta = 2\cos 6\theta - 12\cos 4\theta + 30\cos 2\theta - 20$$

so

$$64\sin^6\theta = 20 - 30\cos 2\theta + 12\cos 4\theta - 2\cos 6\theta$$

and hence

$$\sin^6\theta = \frac{1}{32}\big(10 - 15\cos 2\theta + 6\cos 4\theta - \cos 6\theta\big)$$

as required.

(b) Let

$$I = \int_0^1 \frac{x^6}{(1+x^2)^4}\,\mathrm{d}x$$

Use the substitution

$$x = \tan\theta$$

Then

$$\frac{\mathrm{d}x}{\mathrm{d}\theta} = \sec^2\theta \quad \Longrightarrow \quad \mathrm{d}x = \sec^2\theta\,\mathrm{d}\theta$$

and

$$1 + x^2 = 1 + \tan^2\theta = \sec^2\theta$$

Also, when $x = 0$, $\theta = 0$, and when $x = 1$, $\theta = \frac{\pi}{4}$.

So

$$\begin{aligned} I &= \int_0^{\pi/4} \frac{\tan^6\theta}{(\sec^2\theta)^4}\sec^2\theta\,\mathrm{d}\theta \\ &= \int_0^{\pi/4} \frac{\tan^6\theta}{\sec^6\theta}\,\mathrm{d}\theta \\ &= \int_0^{\pi/4} \frac{\sin^6\theta}{\cos^6\theta}\cos^6\theta\,\mathrm{d}\theta \\ &= \int_0^{\pi/4} \sin^6\theta\,\mathrm{d}\theta \end{aligned}$$

Using the result from part (a):

$$\sin^6\theta = \frac{1}{32}\big(10 - 15\cos 2\theta + 6\cos 4\theta - \cos 6\theta\big)$$

Therefore

$$\begin{aligned} I &= \frac{1}{32}\int_0^{\pi/4}\big(10 - 15\cos 2\theta + 6\cos 4\theta - \cos 6\theta\big)\,\mathrm{d}\theta \\ &= \frac{1}{32}\left[10\theta - \frac{15}{2}\sin 2\theta + \frac{3}{2}\sin 4\theta - \frac{1}{6}\sin 6\theta\right]_0^{\pi/4} \end{aligned}$$

Now substitute the limits:

$$\sin\frac{\pi}{2} = 1, \qquad \sin\pi = 0, \qquad \sin\frac{3\pi}{2} = -1$$

so

$$\begin{aligned} I &= \frac{1}{32}\left(10\times\frac{\pi}{4} - \frac{15}{2}(1) + \frac{3}{2}(0) - \frac{1}{6}(-1)\right) \\ &= \frac{1}{32}\left(\frac{5\pi}{2} - \frac{15}{2} + \frac{1}{6}\right) \\ &= \frac{1}{32}\left(\frac{15\pi - 44}{6}\right) \\ &= \frac{15\pi - 44}{192} \end{aligned}$$

Hence the exact value is

$$\int_0^1 \frac{x^6}{(1+x^2)^4}\,\mathrm{d}x = \frac{15\pi - 44}{192}$$

6. **(a)** Use de Moivre's theorem to show that

$$2\cos 5\theta = 32\cos^5\theta - 40\cos^3\theta + 10\cos\theta$$ **[5]**

(b) Hence solve the equation

$$x^5 - 5x^3 + 5x + 1 = 0$$

giving all roots in the form $2\cos q\pi$, where q is a rational number. **[4]**

Solution

(a) By de Moivre's theorem:

$$(\cos\theta + i\sin\theta)^5 = \cos 5\theta + i\sin 5\theta$$

Expand the left-hand side using the binomial theorem:

$$\begin{aligned}(\cos\theta + i\sin\theta)^5 &= \cos^5\theta + 5i\cos^4\theta\sin\theta + 10i^2\cos^3\theta\sin^2\theta \\ &\quad + 10i^3\cos^2\theta\sin^3\theta + 5i^4\cos\theta\sin^4\theta + i^5\sin^5\theta \\ &= \cos^5\theta + 5i\cos^4\theta\sin\theta - 10\cos^3\theta\sin^2\theta \\ &\quad - 10i\cos^2\theta\sin^3\theta + 5\cos\theta\sin^4\theta + i\sin^5\theta\end{aligned}$$

Equating real parts gives

$$\cos 5\theta = \cos^5\theta - 10\cos^3\theta\sin^2\theta + 5\cos\theta\sin^4\theta$$

Now use

$$\sin^2\theta = 1 - \cos^2\theta \qquad \text{and} \qquad \sin^4\theta = (1-\cos^2\theta)^2$$

So

$$\begin{aligned}\cos 5\theta &= \cos^5\theta - 10\cos^3\theta(1-\cos^2\theta) + 5\cos\theta(1-\cos^2\theta)^2 \\ &= \cos^5\theta - 10\cos^3\theta + 10\cos^5\theta + 5\cos\theta(1 - 2\cos^2\theta + \cos^4\theta) \\ &= \cos^5\theta - 10\cos^3\theta + 10\cos^5\theta + 5\cos\theta - 10\cos^3\theta + 5\cos^5\theta \\ &= 16\cos^5\theta - 20\cos^3\theta + 5\cos\theta\end{aligned}$$

Multiplying by 2:

$$2\cos 5\theta = 32\cos^5\theta - 40\cos^3\theta + 10\cos\theta$$

(b) Let

$$x = 2\cos\theta$$

Then, using part (a):

$$\begin{aligned}x^5 - 5x^3 + 5x &= (2\cos\theta)^5 - 5(2\cos\theta)^3 + 5(2\cos\theta) \\ &= 32\cos^5\theta - 40\cos^3\theta + 10\cos\theta \\ &= 2\cos 5\theta\end{aligned}$$

So the equation

$$x^5 - 5x^3 + 5x + 1 = 0$$

becomes

$$2\cos 5\theta + 1 = 0$$

that is

$$\cos 5\theta = -\frac{1}{2}$$

To obtain distinct values of $x = 2\cos\theta$, take $0 \leq \theta \leq \pi$. Then $0 \leq 5\theta \leq 5\pi$, and in this interval

$$\cos 5\theta = -\frac{1}{2}$$

when

$$5\theta = \frac{2\pi}{3}, \ \frac{4\pi}{3}, \ \frac{8\pi}{3}, \ \frac{10\pi}{3}, \ \frac{14\pi}{3}$$

Hence

$$\theta = \frac{2\pi}{15}, \ \frac{4\pi}{15}, \ \frac{8\pi}{15}, \ \frac{2\pi}{3}, \ \frac{14\pi}{15}$$

Therefore the roots are

$$x = 2\cos\frac{2\pi}{15}, \quad 2\cos\frac{4\pi}{15}, \quad 2\cos\frac{8\pi}{15}, \quad 2\cos\frac{2\pi}{3}, \quad 2\cos\frac{14\pi}{15}$$

These are five distinct roots, so they are all the roots of the equation.

7. **(a)** If n is a positive integer and $z = \cos\theta + i\sin\theta$, show that

$$z^n + \frac{1}{z^n} = 2\cos n\theta, \qquad z^n - \frac{1}{z^n} = 2i\sin n\theta$$

[3]

(b) By considering $\left(z + \frac{1}{z}\right)^3\left(z - \frac{1}{z}\right)^2$, find constants A, B and C such that

$$\sin^2\theta\cos^3\theta = A\cos 5\theta + B\cos 3\theta + C\cos\theta$$

[5]

Solution

(a) Write

$$z = \cos\theta + i\sin\theta$$

Then, by de Moivre's theorem:

$$z^n = \cos n\theta + i\sin n\theta$$

and

$$\frac{1}{z^n} = z^{-n} = \cos(-n\theta) + i\sin(-n\theta) = \cos n\theta - i\sin n\theta$$

Adding gives

$$\begin{aligned} z^n + \frac{1}{z^n} &= (\cos n\theta + i\sin n\theta) + (\cos n\theta - i\sin n\theta) \\ &= 2\cos n\theta \end{aligned}$$

Subtracting gives

$$\begin{aligned} z^n - \frac{1}{z^n} &= (\cos n\theta + i\sin n\theta) - (\cos n\theta - i\sin n\theta) \\ &= 2i\sin n\theta \end{aligned}$$

Hence

$$z^n + \frac{1}{z^n} = 2\cos n\theta, \qquad z^n - \frac{1}{z^n} = 2i\sin n\theta$$

(b) From part (a), with $n = 1$:

$$z + \frac{1}{z} = 2\cos\theta, \qquad z - \frac{1}{z} = 2i\sin\theta$$

So

$$\begin{aligned} \left(z + \frac{1}{z}\right)^3\left(z - \frac{1}{z}\right)^2 &= (2\cos\theta)^3(2i\sin\theta)^2 \\ &= 8\cos^3\theta \times 4i^2\sin^2\theta \\ &= 8\cos^3\theta \times (-4)\sin^2\theta \\ &= -32\sin^2\theta\cos^3\theta \end{aligned}$$

Now expand the expression algebraically.

First

$$\left(z+\frac{1}{z}\right)^3 = z^3+3z+\frac{3}{z}+\frac{1}{z^3}$$

and

$$\left(z-\frac{1}{z}\right)^2 = z^2-2+\frac{1}{z^2}$$

Hence

$$\begin{aligned}\left(z+\frac{1}{z}\right)^3\left(z-\frac{1}{z}\right)^2 &= \left(z^3+3z+\frac{3}{z}+\frac{1}{z^3}\right)\left(z^2-2+\frac{1}{z^2}\right)\\ &= \left(z^5-2z^3+z\right)+\left(3z^3-6z+\frac{3}{z}\right)\\ &\quad +\left(3z-\frac{6}{z}+\frac{3}{z^3}\right)+\left(\frac{1}{z}-\frac{2}{z^3}+\frac{1}{z^5}\right)\\ &= z^5+z^3-2z-\frac{2}{z}+\frac{1}{z^3}+\frac{1}{z^5}\end{aligned}$$

Group the terms in pairs:

$$\left(z^5+\frac{1}{z^5}\right)+\left(z^3+\frac{1}{z^3}\right)-2\left(z+\frac{1}{z}\right)$$

Using part (a) again:

$$z^5+\frac{1}{z^5}=2\cos 5\theta, \qquad z^3+\frac{1}{z^3}=2\cos 3\theta, \qquad z+\frac{1}{z}=2\cos\theta$$

So

$$\left(z+\frac{1}{z}\right)^3\left(z-\frac{1}{z}\right)^2 = 2\cos 5\theta+2\cos 3\theta-4\cos\theta$$

Equating the two expressions:

$$-32\sin^2\theta\cos^3\theta = 2\cos 5\theta+2\cos 3\theta-4\cos\theta$$

Therefore

$$\begin{aligned}\sin^2\theta\cos^3\theta &= -\frac{1}{32}(2\cos 5\theta+2\cos 3\theta-4\cos\theta)\\ &= -\frac{1}{16}\cos 5\theta-\frac{1}{16}\cos 3\theta+\frac{1}{8}\cos\theta\end{aligned}$$

So the constants are

$$A=-\frac{1}{16}, \qquad B=-\frac{1}{16}, \qquad C=\frac{1}{8}$$

8. **(a)** By writing $1 + i\tan\theta = \sec\theta(\cos\theta + i\sin\theta)$ and using de Moivre's theorem, show that

$$(1 + i\tan\theta)^4 = \frac{\cos 4\theta + i\sin 4\theta}{\cos^4\theta}$$

[3]

(b) Expand $(1 + i\tan\theta)^4$ and hence show that

$$\cos 4\theta = \frac{1 - 6\tan^2\theta + \tan^4\theta}{(1 + \tan^2\theta)^2}$$

and

$$\sin 4\theta = \frac{4\tan\theta(1 - \tan^2\theta)}{(1 + \tan^2\theta)^2}$$

[4]

(c) Deduce that

$$\tan 4\theta = \frac{4\tan\theta(1 - \tan^2\theta)}{1 - 6\tan^2\theta + \tan^4\theta}$$

[2]

Solution

(a) Starting with

$$1 + i\tan\theta = 1 + i\frac{\sin\theta}{\cos\theta} = \frac{\cos\theta + i\sin\theta}{\cos\theta} = \sec\theta(\cos\theta + i\sin\theta)$$

Raise both sides to the 4th power:

$$(1 + i\tan\theta)^4 = \sec^4\theta(\cos\theta + i\sin\theta)^4$$

By de Moivre's theorem:

$$(\cos\theta + i\sin\theta)^4 = \cos 4\theta + i\sin 4\theta$$

so

$$(1 + i\tan\theta)^4 = \sec^4\theta(\cos 4\theta + i\sin 4\theta) = \frac{\cos 4\theta + i\sin 4\theta}{\cos^4\theta}$$

(b) Let $t = \tan\theta$. Then

$$(1 + i\tan\theta)^4 = (1 + it)^4$$

Expanding:

$$\begin{aligned}(1 + it)^4 &= 1 + 4it + 6(it)^2 + 4(it)^3 + (it)^4 \\ &= 1 + 4it + 6i^2t^2 + 4i^3t^3 + i^4t^4 \\ &= 1 + 4it - 6t^2 - 4it^3 + t^4 \\ &= (1 - 6t^2 + t^4) + i(4t - 4t^3) \\ &= (1 - 6t^2 + t^4) + i\,4t(1 - t^2)\end{aligned}$$

From part (a):

$$(1+it)^4 = \frac{\cos 4\theta + i \sin 4\theta}{\cos^4 \theta}$$

Equating real and imaginary parts gives

$$\frac{\cos 4\theta}{\cos^4 \theta} = 1 - 6t^2 + t^4 \qquad \text{and} \qquad \frac{\sin 4\theta}{\cos^4 \theta} = 4t(1-t^2)$$

Now

$$1 + \tan^2 \theta = \sec^2 \theta = \frac{1}{\cos^2 \theta}$$

so

$$\cos^4 \theta = \frac{1}{(1+\tan^2 \theta)^2} = \frac{1}{(1+t^2)^2}$$

Therefore

$$\begin{aligned} \cos 4\theta &= (1 - 6t^2 + t^4)\cos^4 \theta \\ &= \frac{1 - 6t^2 + t^4}{(1+t^2)^2} \end{aligned}$$

and

$$\begin{aligned} \sin 4\theta &= 4t(1-t^2)\cos^4 \theta \\ &= \frac{4t(1-t^2)}{(1+t^2)^2} \end{aligned}$$

Replacing t by $\tan \theta$ gives

$$\cos 4\theta = \frac{1 - 6\tan^2 \theta + \tan^4 \theta}{(1+\tan^2 \theta)^2}$$

and

$$\sin 4\theta = \frac{4\tan \theta(1 - \tan^2 \theta)}{(1+\tan^2 \theta)^2}$$

(c) Using

$$\tan 4\theta = \frac{\sin 4\theta}{\cos 4\theta}$$

and the results from part (b):

$$\begin{aligned} \tan 4\theta &= \frac{\dfrac{4\tan \theta(1 - \tan^2 \theta)}{(1+\tan^2 \theta)^2}}{\dfrac{1 - 6\tan^2 \theta + \tan^4 \theta}{(1+\tan^2 \theta)^2}} \\ &= \frac{4\tan \theta(1 - \tan^2 \theta)}{1 - 6\tan^2 \theta + \tan^4 \theta} \end{aligned}$$

9. **(a)** Use de Moivre's theorem to show that

$$\sec 6\theta = \frac{\sec^6\theta}{32 - 48\sec^2\theta + 18\sec^4\theta - \sec^6\theta}$$ **[6]**

(b) Hence obtain the roots of the equation

$$x^6 - 18x^4 + 48x^2 - 32 = 0$$

in the form $\sec(q\pi)$, where q is rational. **[4]**

Solution

(a) By de Moivre's theorem:

$$(\cos\theta + i\sin\theta)^6 = \cos 6\theta + i\sin 6\theta$$

Let $c = \cos\theta$ and $s = \sin\theta$. Expanding the left-hand side:

$$(c+is)^6 = c^6 + 6c^5(is) + 15c^4(is)^2 + 20c^3(is)^3 + 15c^2(is)^4 + 6c(is)^5 + (is)^6$$

Equating real parts gives

$$\cos 6\theta = c^6 - 15c^4s^2 + 15c^2s^4 - s^6$$

Now use $s^2 = 1 - c^2$:

$$\begin{aligned}\cos 6\theta &= c^6 - 15c^4(1-c^2) + 15c^2(1-c^2)^2 - (1-c^2)^3\\ &= c^6 - 15c^4 + 15c^6 + 15c^2(1 - 2c^2 + c^4) - (1 - 3c^2 + 3c^4 - c^6)\\ &= c^6 - 15c^4 + 15c^6 + 15c^2 - 30c^4 + 15c^6 - 1 + 3c^2 - 3c^4 + c^6\\ &= 32c^6 - 48c^4 + 18c^2 - 1\end{aligned}$$

So

$$\cos 6\theta = 32\cos^6\theta - 48\cos^4\theta + 18\cos^2\theta - 1$$

Hence

$$\sec 6\theta = \frac{1}{32\cos^6\theta - 48\cos^4\theta + 18\cos^2\theta - 1}$$

Multiply top and bottom by $\sec^6\theta$:

$$\sec 6\theta = \frac{\sec^6\theta}{32 - 48\sec^2\theta + 18\sec^4\theta - \sec^6\theta}$$

(b) Let

$$x = \sec\theta$$

Then the equation

$$x^6 - 18x^4 + 48x^2 - 32 = 0$$

becomes

$$\sec^6\theta - 18\sec^4\theta + 48\sec^2\theta - 32 = 0$$

or

$$32 - 48\sec^2\theta + 18\sec^4\theta - \sec^6\theta = 0$$

From part (a):

$$\sec 6\theta = \frac{\sec^6\theta}{32 - 48\sec^2\theta + 18\sec^4\theta - \sec^6\theta}$$

So if

$$32 - 48\sec^2\theta + 18\sec^4\theta - \sec^6\theta = 0$$

then $\sec 6\theta$ is undefined. Since $\sec^6\theta \neq 0$, this means

$$\cos 6\theta = 0$$

Therefore

$$6\theta = \frac{(2k+1)\pi}{2} \qquad (k \in \mathbb{Z})$$

so

$$\theta = \frac{(2k+1)\pi}{12}$$

Hence

$$x = \sec\left(\frac{(2k+1)\pi}{12}\right)$$

Taking $k = 0, 1, 2, 3, 4, 5$ gives the six distinct roots:

$$\sec\frac{\pi}{12}, \quad \sec\frac{\pi}{4}, \quad \sec\frac{5\pi}{12}, \quad \sec\frac{7\pi}{12}, \quad \sec\frac{3\pi}{4}, \quad \sec\frac{11\pi}{12}$$

10. **(a)** Use de Moivre's theorem to show that

$$\cos 5\theta = 16\cos^5\theta - 20\cos^3\theta + 5\cos\theta$$ **[5]**

(b) Hence solve for $0 \leq \theta \leq \pi$

$$16\cos^5\theta - 20\cos^3\theta + 5\cos\theta = \cos 2\theta$$

giving your answers as exact multiples of π. **[5]**

Solution

(a) Let

$$z = \cos\theta + i\sin\theta$$

By de Moivre's theorem:

$$z^5 = \cos 5\theta + i\sin 5\theta$$

Now expand z^5 using the binomial theorem:

$$\begin{aligned}(\cos\theta + i\sin\theta)^5 &= \cos^5\theta + 5\cos^4\theta(i\sin\theta) + 10\cos^3\theta(i\sin\theta)^2 \\ &\quad + 10\cos^2\theta(i\sin\theta)^3 + 5\cos\theta(i\sin\theta)^4 + (i\sin\theta)^5 \\ &= \cos^5\theta + 5i\cos^4\theta\sin\theta - 10\cos^3\theta\sin^2\theta \\ &\quad - 10i\cos^2\theta\sin^3\theta + 5\cos\theta\sin^4\theta + i\sin^5\theta\end{aligned}$$

Equating real parts gives

$$\cos 5\theta = \cos^5\theta - 10\cos^3\theta\sin^2\theta + 5\cos\theta\sin^4\theta$$

Now use

$$\sin^2\theta = 1 - \cos^2\theta, \qquad \sin^4\theta = (1-\cos^2\theta)^2$$

So

$$\begin{aligned}\cos 5\theta &= \cos^5\theta - 10\cos^3\theta(1-\cos^2\theta) + 5\cos\theta(1-\cos^2\theta)^2 \\ &= \cos^5\theta - 10\cos^3\theta + 10\cos^5\theta + 5\cos\theta(1 - 2\cos^2\theta + \cos^4\theta) \\ &= \cos^5\theta - 10\cos^3\theta + 10\cos^5\theta + 5\cos\theta - 10\cos^3\theta + 5\cos^5\theta \\ &= 16\cos^5\theta - 20\cos^3\theta + 5\cos\theta\end{aligned}$$

(b) Using part (a), the equation

$$16\cos^5\theta - 20\cos^3\theta + 5\cos\theta = \cos 2\theta$$

becomes

$$\cos 5\theta = \cos 2\theta$$

For $\cos A = \cos B$, the general solution is

$$A = 2n\pi \pm B \qquad (n \in \mathbb{Z})$$

So

$$5\theta = 2n\pi \pm 2\theta$$

This gives two cases.

From

$$5\theta = 2n\pi + 2\theta$$

we get

$$3\theta = 2n\pi$$

so

$$\theta = \frac{2n\pi}{3}$$

For $0 \leq \theta \leq \pi$, this gives

$$\theta = 0,\ \frac{2\pi}{3}$$

From

$$5\theta = 2n\pi - 2\theta$$

we get

$$7\theta = 2n\pi$$

so

$$\theta = \frac{2n\pi}{7}$$

For $0 \leq \theta \leq \pi$, this gives

$$\theta = 0,\ \frac{2\pi}{7},\ \frac{4\pi}{7},\ \frac{6\pi}{7}$$

Combining the distinct solutions:

$$\theta = 0,\ \frac{2\pi}{7},\ \frac{4\pi}{7},\ \frac{2\pi}{3},\ \frac{6\pi}{7}$$

11. **(a)** Use de Moivre's theorem to show that $\cos 6\theta \equiv (1-2\sin^2\theta)(16\sin^4\theta - 16\sin^2\theta + 1)$ **[5]**

(b) By solving the equation $\cos 6\theta = 0$, deduce that $\sin^2\left(\frac{\pi}{12}\right) = \frac{2-\sqrt{3}}{4}$. **[4]**

(c) Hence, or otherwise, write down the exact values of $\sin^2\left(\frac{5\pi}{12}\right)$, $\sin^2\left(\frac{7\pi}{12}\right)$ and $\sin^2\left(\frac{11\pi}{12}\right)$. **[3]**

Solution

(a) Using de Moivre's theorem:

$$(\cos\theta + i\sin\theta)^6 = \cos 6\theta + i\sin 6\theta$$

Expand the left-hand side by the binomial theorem:

$$\begin{aligned}(\cos\theta + i\sin\theta)^6 &= \cos^6\theta + 6\cos^5\theta(i\sin\theta) + 15\cos^4\theta(i\sin\theta)^2 \\ &\quad + 20\cos^3\theta(i\sin\theta)^3 + 15\cos^2\theta(i\sin\theta)^4 \\ &\quad + 6\cos\theta(i\sin\theta)^5 + (i\sin\theta)^6 \\ &= \cos^6\theta + 6i\cos^5\theta\sin\theta - 15\cos^4\theta\sin^2\theta \\ &\quad - 20i\cos^3\theta\sin^3\theta + 15\cos^2\theta\sin^4\theta \\ &\quad + 6i\cos\theta\sin^5\theta - \sin^6\theta\end{aligned}$$

Equating real parts gives

$$\cos 6\theta = \cos^6\theta - 15\cos^4\theta\sin^2\theta + 15\cos^2\theta\sin^4\theta - \sin^6\theta$$

Now let

$$u = \sin^2\theta \qquad \text{so} \qquad \cos^2\theta = 1 - u$$

Then

$$\begin{aligned}\cos 6\theta &= (1-u)^3 - 15(1-u)^2u + 15(1-u)u^2 - u^3 \\ &= (1 - 3u + 3u^2 - u^3) - 15(u - 2u^2 + u^3) + 15(u^2 - u^3) - u^3 \\ &= 1 - 3u + 3u^2 - u^3 - 15u + 30u^2 - 15u^3 + 15u^2 - 15u^3 - u^3 \\ &= 1 - 18u + 48u^2 - 32u^3\end{aligned}$$

Now factorise:

$$1 - 18u + 48u^2 - 32u^3 = (1-2u)(16u^2 - 16u + 1)$$

Substituting back $u = \sin^2\theta$:

$$\cos 6\theta = (1-2\sin^2\theta)(16\sin^4\theta - 16\sin^2\theta + 1)$$

Hence

$$\cos 6\theta \equiv (1-2\sin^2\theta)(16\sin^4\theta - 16\sin^2\theta + 1)$$

(b) Let

$$u = \sin^2\theta$$

From part (a), if $\cos 6\theta = 0$, then

$$(1-2u)(16u^2-16u+1)=0$$

So either

$$1-2u=0 \quad \Longrightarrow \quad u=\frac{1}{2}$$

or

$$16u^2-16u+1=0$$

$$\begin{aligned} u &= \frac{16 \pm \sqrt{(-16)^2-4(16)(1)}}{2\cdot 16} \\ &= \frac{16 \pm \sqrt{256-64}}{32} \\ &= \frac{16 \pm \sqrt{192}}{32} \\ &= \frac{16 \pm 8\sqrt{3}}{32} \\ &= \frac{2 \pm \sqrt{3}}{4} \end{aligned}$$

Now

$$\cos\left(6 \times \frac{\pi}{12}\right) = \cos\frac{\pi}{2} = 0$$

so $\sin^2\left(\frac{\pi}{12}\right)$ must be one of

$$\frac{1}{2}, \quad \frac{2+\sqrt{3}}{4}, \quad \frac{2-\sqrt{3}}{4}$$

But

$$0 < \frac{\pi}{12} < \frac{\pi}{4}$$

so

$$\sin^2\left(\frac{\pi}{12}\right) < \sin^2\left(\frac{\pi}{4}\right) = \frac{1}{2}$$

Therefore the correct value is

$$\sin^2\left(\frac{\pi}{12}\right) = \frac{2-\sqrt{3}}{4}$$

(c) Using part (b):

$$\begin{aligned} \sin^2\left(\frac{5\pi}{12}\right) &= \cos^2\left(\frac{\pi}{12}\right) \\ &= 1-\sin^2\left(\frac{\pi}{12}\right) \\ &= 1-\frac{2-\sqrt{3}}{4} \\ &= \frac{2+\sqrt{3}}{4} \end{aligned}$$

Also

$$\sin^2\left(\frac{7\pi}{12}\right) = \sin^2\left(\pi - \frac{5\pi}{12}\right) = \sin^2\left(\frac{5\pi}{12}\right) = \frac{2+\sqrt{3}}{4}$$

and

$$\sin^2\left(\frac{11\pi}{12}\right) = \sin^2\left(\pi - \frac{\pi}{12}\right) = \sin^2\left(\frac{\pi}{12}\right) = \frac{2-\sqrt{3}}{4}$$

Hence

$$\sin^2\left(\frac{5\pi}{12}\right) = \frac{2+\sqrt{3}}{4}, \qquad \sin^2\left(\frac{7\pi}{12}\right) = \frac{2+\sqrt{3}}{4}, \qquad \sin^2\left(\frac{11\pi}{12}\right) = \frac{2-\sqrt{3}}{4}$$

12. **(i)** Use de Moivre's theorem to show that

$$\cos 5\theta \equiv \cos\theta\left(1 - 12\sin^2\theta + 16\sin^4\theta\right)$$ **[5]**

(ii) Hence determine the range of values of

$$\frac{\cos 5\theta}{\cos\theta}$$

given that $\cos\theta \neq 0$. **[7]**

Solution

(i) Using de Moivre's theorem:

$$(\cos\theta + i\sin\theta)^5 = \cos 5\theta + i\sin 5\theta$$

Expand the left-hand side by the binomial theorem:

$$\begin{aligned}(\cos\theta + i\sin\theta)^5 = \cos^5\theta + 5\cos^4\theta(i\sin\theta) + 10\cos^3\theta(i\sin\theta)^2 \\ + 10\cos^2\theta(i\sin\theta)^3 + 5\cos\theta(i\sin\theta)^4 + (i\sin\theta)^5\end{aligned}$$

Now use

$$i^2 = -1, \qquad i^3 = -i, \qquad i^4 = 1, \qquad i^5 = i$$

So

$$\begin{aligned}(\cos\theta + i\sin\theta)^5 = \cos^5\theta + 5i\cos^4\theta\sin\theta - 10\cos^3\theta\sin^2\theta \\ - 10i\cos^2\theta\sin^3\theta + 5\cos\theta\sin^4\theta + i\sin^5\theta\end{aligned}$$

Equating real parts gives

$$\cos 5\theta = \cos^5\theta - 10\cos^3\theta\sin^2\theta + 5\cos\theta\sin^4\theta$$

Factor out $\cos\theta$:

$$\cos 5\theta = \cos\theta\left(\cos^4\theta - 10\cos^2\theta\sin^2\theta + 5\sin^4\theta\right)$$

Now use $\cos^2\theta = 1 - \sin^2\theta$:

$$\begin{aligned}\cos 5\theta &= \cos\theta\left((1 - \sin^2\theta)^2 - 10(1 - \sin^2\theta)\sin^2\theta + 5\sin^4\theta\right) \\ &= \cos\theta\left(1 - 2\sin^2\theta + \sin^4\theta - 10\sin^2\theta + 10\sin^4\theta + 5\sin^4\theta\right) \\ &= \cos\theta\left(1 - 12\sin^2\theta + 16\sin^4\theta\right)\end{aligned}$$

Hence

$$\cos 5\theta \equiv \cos\theta\left(1 - 12\sin^2\theta + 16\sin^4\theta\right)$$

(ii) From part (i):

$$\frac{\cos 5\theta}{\cos\theta} = 1 - 12\sin^2\theta + 16\sin^4\theta$$

provided $\cos\theta \neq 0$.

Let

$$u = \sin^2\theta$$

Then, since $0 \le \sin^2\theta \le 1$ and $\cos\theta \neq 0$, we have

$$0 \le u < 1$$

So we need the range of

$$f(u) = 1 - 12u + 16u^2 \qquad \text{for } 0 \le u < 1$$

The coefficient of u^2 is positive, so the minimum is at the turning point, where

$$u = \frac{-(-12)}{2 \times 16} = \frac{12}{32} = \frac{3}{8}$$

The minimum value is

$$\begin{aligned} f\left(\frac{3}{8}\right) &= 1 - 12\left(\frac{3}{8}\right) + 16\left(\frac{3}{8}\right)^2 \\ &= 1 - \frac{36}{8} + 16 \times \frac{9}{64} \\ &= 1 - \frac{9}{2} + \frac{9}{4} \\ &= \frac{4}{4} - \frac{18}{4} + \frac{9}{4} \\ &= -\frac{5}{4} \end{aligned}$$

So the least possible value is

$$-\frac{5}{4}$$

Now consider the largest possible value on $0 \le u < 1$.

As $u \to 1^-$:

$$f(u) \to 1 - 12 + 16 = 5$$

We must check whether 5 is actually attained. Solve

$$\begin{aligned} 1 - 12u + 16u^2 &= 5 \\ 16u^2 - 12u - 4 &= 0 \\ 4u^2 - 3u - 1 &= 0 \\ (4u + 1)(u - 1) &= 0 \end{aligned}$$

So

$$u = 1 \quad \text{or} \quad u = -\frac{1}{4}$$

But $u = \sin^2\theta$ cannot be negative, and $u = 1$ would mean

$$\sin^2\theta = 1 \quad \Longrightarrow \quad \cos\theta = 0$$

which is excluded.

So 5 is not attained.

Therefore the range of

$$\frac{\cos 5\theta}{\cos \theta}$$

is

$$-\frac{5}{4} \leq \frac{\cos 5\theta}{\cos \theta} < 5$$

13. **(a)** Use de Moivre's theorem to show that

$$\cos 5\theta = 16\cos^5\theta - 20\cos^3\theta + 5\cos\theta$$ **[5]**

(b) Use the identity given in part (a) to find the 2 positive roots of

$$64x^5 - 80x^3 + 20x + 1 = 0$$

giving your answers to 3 significant figures. **[4]**

Solution

(a) Let $c = \cos\theta$ and $s = \sin\theta$. By de Moivre's theorem:

$$(c + is)^5 = \cos 5\theta + i\sin 5\theta$$

Expanding the left-hand side:

$$\begin{aligned}(c+is)^5 &= c^5 + 5c^4(is) + 10c^3(is)^2 + 10c^2(is)^3 + 5c(is)^4 + (is)^5 \\ &= c^5 + 5ic^4s - 10c^3s^2 - 10ic^2s^3 + 5cs^4 + is^5\end{aligned}$$

Equating real parts gives

$$\cos 5\theta = c^5 - 10c^3s^2 + 5cs^4$$

Now use $s^2 = 1 - c^2$ and $s^4 = (1-c^2)^2$:

$$\begin{aligned}\cos 5\theta &= c^5 - 10c^3(1-c^2) + 5c(1-c^2)^2 \\ &= c^5 - 10c^3 + 10c^5 + 5c(1 - 2c^2 + c^4) \\ &= c^5 - 10c^3 + 10c^5 + 5c - 10c^3 + 5c^5 \\ &= 16c^5 - 20c^3 + 5c\end{aligned}$$

Since $c = \cos\theta$, this is

$$\cos 5\theta = 16\cos^5\theta - 20\cos^3\theta + 5\cos\theta$$

as required.

(b) Let

$$x = \cos\theta \qquad \text{with } 0^\circ \leq \theta \leq 180^\circ$$

Then

$$64x^5 - 80x^3 + 20x + 1 = 0$$

$$4(16x^5 - 20x^3 + 5x) + 1 = 0$$

Using the identity from part (a):

$$16x^5 - 20x^3 + 5x = \cos 5\theta$$

so

$$4\cos 5\theta + 1 = 0$$

Hence

$$\cos 5\theta = -\frac{1}{4}$$

Let

$$\alpha = \arccos\left(-\frac{1}{4}\right) \approx 104.478^\circ$$

Since $0^\circ \leq \theta \leq 180^\circ$, we have $0^\circ \leq 5\theta \leq 900^\circ$. Therefore the solutions for 5θ in this interval are

$$\alpha, \quad 360^\circ - \alpha, \quad 360^\circ + \alpha, \quad 720^\circ - \alpha, \quad 720^\circ + \alpha$$

So

$$\theta \approx 20.896^\circ,\ 51.104^\circ,\ 92.896^\circ,\ 123.104^\circ,\ 164.896^\circ$$

Now $x = \cos\theta$, and $x > 0$ when $0^\circ < \theta < 90^\circ$. So only the first two values give positive roots:

$$x = \cos 20.896^\circ \approx 0.934 \qquad (3 \text{ s.f.})$$

$$x = \cos 51.104^\circ \approx 0.628 \qquad (3 \text{ s.f.})$$

So the two positive roots are $x \approx 0.628$ and $x \approx 0.934$.

14. **(a)** Use de Moivre's theorem to show that

$$\sin^2\theta\cos^2\theta = \frac{1}{8}(1-\cos 4\theta)$$

[5]

(b) Hence, or otherwise, find the solution of the differential equation

$$\frac{\mathrm{d}y}{\mathrm{d}\theta} - 2y\tan\theta = \sin^2\theta$$

for which $y = 0$ when $\theta = \frac{1}{4}\pi$. **[6]**

Solution

(a) By de Moivre's theorem:

$$(\cos\theta + i\sin\theta)^4 = \cos 4\theta + i\sin 4\theta$$

Expanding the left-hand side:

$$\begin{aligned}(\cos\theta + i\sin\theta)^4 &= \cos^4\theta + 4\cos^3\theta(i\sin\theta) + 6\cos^2\theta(i\sin\theta)^2 \\ &\quad + 4\cos\theta(i\sin\theta)^3 + (i\sin\theta)^4 \\ &= \cos^4\theta + 4i\cos^3\theta\sin\theta - 6\cos^2\theta\sin^2\theta \\ &\quad - 4i\cos\theta\sin^3\theta + \sin^4\theta\end{aligned}$$

Equating real parts gives

$$\cos 4\theta = \cos^4\theta - 6\sin^2\theta\cos^2\theta + \sin^4\theta$$

Now rewrite the right-hand side as

$$\begin{aligned}\cos^4\theta - 6\sin^2\theta\cos^2\theta + \sin^4\theta &= (\sin^2\theta + \cos^2\theta)^2 - 8\sin^2\theta\cos^2\theta \\ &= 1 - 8\sin^2\theta\cos^2\theta\end{aligned}$$

Hence

$$\cos 4\theta = 1 - 8\sin^2\theta\cos^2\theta$$

Therefore

$$\sin^2\theta\cos^2\theta = \frac{1-\cos 4\theta}{8}$$

So

$$\sin^2\theta\cos^2\theta = \frac{1}{8}(1-\cos 4\theta)$$

(b) The differential equation is

$$\frac{\mathrm{d}y}{\mathrm{d}\theta} - 2y\tan\theta = \sin^2\theta$$

This is a linear first order differential equation of the form

$$\frac{\mathrm{d}y}{\mathrm{d}\theta} + P(\theta)y = Q(\theta)$$

with

$$P(\theta) = -2\tan\theta$$

The integrating factor is

$$\mu(\theta) = e^{\int -2\tan\theta\,\mathrm{d}\theta}$$

Using

$$\int \tan\theta\,\mathrm{d}\theta = \ln|\sec\theta|$$

we get

$$\mu(\theta) = e^{-2\ln|\sec\theta|} = \cos^2\theta$$

Multiply the differential equation through by $\cos^2\theta$:

$$\cos^2\theta\frac{\mathrm{d}y}{\mathrm{d}\theta} - 2y\sin\theta\cos\theta = \sin^2\theta\cos^2\theta$$

The left-hand side is the derivative of $y\cos^2\theta$, so

$$\frac{\mathrm{d}}{\mathrm{d}\theta}(y\cos^2\theta) = \sin^2\theta\cos^2\theta$$

Using part (a):

$$\frac{\mathrm{d}}{\mathrm{d}\theta}(y\cos^2\theta) = \frac{1-\cos 4\theta}{8}$$

Integrating:

$$\begin{aligned} y\cos^2\theta &= \int \frac{1-\cos 4\theta}{8}\,\mathrm{d}\theta \\ &= \frac{1}{8}\int 1\,\mathrm{d}\theta - \frac{1}{8}\int \cos 4\theta\,\mathrm{d}\theta \\ &= \frac{\theta}{8} - \frac{\sin 4\theta}{32} + c \end{aligned}$$

Now use the condition $y = 0$ when $\theta = \frac{\pi}{4}$:

$$0 \times \cos^2\frac{\pi}{4} = \frac{1}{8}\times\frac{\pi}{4} - \frac{\sin\pi}{32} + c$$

So

$$0 = \frac{\pi}{32} + c$$

and hence

$$c = -\frac{\pi}{32}$$

Therefore

$$y\cos^2\theta = \frac{\theta}{8} - \frac{\sin 4\theta}{32} - \frac{\pi}{32}$$

So

$$y = \frac{\frac{4\theta}{32} - \frac{\sin 4\theta}{32} - \frac{\pi}{32}}{\cos^2\theta} = \frac{4\theta - \sin 4\theta - \pi}{32\cos^2\theta}$$

15. **(a)** Use de Moivre's theorem to show that

$$\cos^6\theta = a\cos 6\theta + b\cos 4\theta + c\cos 2\theta + d$$

where a, b, c and d are constants to be found. **[5]**

(b) Hence show that

$$\int_0^{\pi/4} \cos^6\theta \,\mathrm{d}\theta = \frac{11}{48} + \frac{5\pi}{64}$$ **[4]**

Solution

(a) Let $z = \cos\theta + i\sin\theta$. By de Moivre's theorem:

$$z^n + z^{-n} = (\cos n\theta + i\sin n\theta) + (\cos n\theta - i\sin n\theta) = 2\cos n\theta$$

In particular, $z + z^{-1} = 2\cos\theta$, so

$$(z + z^{-1})^6 = 64\cos^6\theta$$

Expanding by the binomial theorem:

$$(z + z^{-1})^6 = z^6 + 6z^4 + 15z^2 + 20 + 15z^{-2} + 6z^{-4} + z^{-6}$$

Group the terms in pairs:

$$\begin{aligned} 64\cos^6\theta &= (z^6 + z^{-6}) + 6(z^4 + z^{-4}) + 15(z^2 + z^{-2}) + 20 \\ &= 2\cos 6\theta + 12\cos 4\theta + 30\cos 2\theta + 20 \end{aligned}$$

Dividing by 64:

$$\begin{aligned} \cos^6\theta &= \frac{1}{32}\cos 6\theta + \frac{12}{64}\cos 4\theta + \frac{30}{64}\cos 2\theta + \frac{20}{64} \\ &= \frac{1}{32}\cos 6\theta + \frac{3}{16}\cos 4\theta + \frac{15}{32}\cos 2\theta + \frac{5}{16} \end{aligned}$$

Hence

$$a = \frac{1}{32}, \qquad b = \frac{3}{16}, \qquad c = \frac{15}{32}, \qquad d = \frac{5}{16}$$

(b) Using the identity from part (a):

$$\cos^6\theta = \frac{1}{32}\big(\cos 6\theta + 6\cos 4\theta + 15\cos 2\theta + 10\big)$$

So

$$\begin{aligned} \int_0^{\pi/4} \cos^6\theta \,\mathrm{d}\theta &= \frac{1}{32}\int_0^{\pi/4} \big(\cos 6\theta + 6\cos 4\theta + 15\cos 2\theta + 10\big)\,\mathrm{d}\theta \\ &= \frac{1}{32}\left[\frac{1}{6}\sin 6\theta + \frac{6}{4}\sin 4\theta + \frac{15}{2}\sin 2\theta + 10\theta\right]_0^{\pi/4} \\ &= \frac{1}{32}\left[\frac{1}{6}\sin 6\theta + \frac{3}{2}\sin 4\theta + \frac{15}{2}\sin 2\theta + 10\theta\right]_0^{\pi/4} \end{aligned}$$

At $\theta = \frac{\pi}{4}$:

$$\sin\frac{3\pi}{2} = -1, \qquad \sin\pi = 0, \qquad \sin\frac{\pi}{2} = 1$$

Therefore

$$\begin{aligned}
\int_0^{\pi/4} \cos^6\theta \,\mathrm{d}\theta &= \frac{1}{32}\left(-\frac{1}{6} + \frac{3}{2}\times 0 + \frac{15}{2}\times 1 + 10\times\frac{\pi}{4}\right) \\
&= \frac{1}{32}\left(-\frac{1}{6} + \frac{15}{2} + \frac{5\pi}{2}\right) \\
&= \frac{1}{32}\left(\frac{44}{6} + \frac{5\pi}{2}\right) \\
&= \frac{11}{48} + \frac{5\pi}{64}
\end{aligned}$$

16. **(i)** Starting with the result

$$e^{i\theta} = \cos\theta + i\sin\theta$$

show that

(A)

$$(\cos\theta + i\sin\theta)^n = \cos n\theta + i\sin n\theta \quad \textbf{[2]}$$

(B)

$$\sin\theta = \frac{1}{2i}\left(e^{i\theta} - e^{-i\theta}\right) \quad \textbf{[2]}$$

(ii) Using the result in part (i)(A), obtain the values of the constants a, b, c, d and e in the identity

$$\cos 8\theta \equiv a\sin^8\theta + b\sin^6\theta + c\sin^4\theta + d\sin^2\theta + e \quad \textbf{[6]}$$

(iii) Using the result in part (i)(B), obtain the values of the constants P, Q, R, S and T in the identity

$$\sin^8\theta \equiv P\cos 8\theta + Q\cos 6\theta + R\cos 4\theta + S\cos 2\theta + T \quad \textbf{[6]}$$

(iv) Hence show that

$$\sin\frac{\pi}{8} = \left(\frac{17 - 12\sqrt{2}}{64}\right)^{\frac{1}{8}} \quad \textbf{[3]}$$

Solution

(i) (A) Starting from

$$e^{i\theta} = \cos\theta + i\sin\theta$$

raise both sides to the power n:

$$(\cos\theta + i\sin\theta)^n = \left(e^{i\theta}\right)^n = e^{in\theta}$$

Now apply Euler's formula again, with θ replaced by $n\theta$:

$$e^{in\theta} = \cos n\theta + i\sin n\theta$$

Hence

$$(\cos\theta + i\sin\theta)^n = \cos n\theta + i\sin n\theta$$

(B) Replace θ by $-\theta$ in Euler's formula:

$$e^{-i\theta} = \cos(-\theta) + i\sin(-\theta) = \cos\theta - i\sin\theta$$

Subtract this from

$$e^{i\theta} = \cos\theta + i\sin\theta$$

to get

$$e^{i\theta} - e^{-i\theta} = 2i\sin\theta$$

Therefore

$$\sin\theta = \frac{1}{2i}\left(e^{i\theta} - e^{-i\theta}\right)$$

(ii) From part (i)(A):

$$(\cos\theta + i\sin\theta)^8 = \cos 8\theta + i\sin 8\theta$$

Expand $(\cos\theta + i\sin\theta)^8$:

$$\begin{aligned}(\cos\theta + i\sin\theta)^8 &= \cos^8\theta + 8i\cos^7\theta\sin\theta + 28\cos^6\theta(i\sin\theta)^2 + 56\cos^5\theta(i\sin\theta)^3 \\ &\quad + 70\cos^4\theta(i\sin\theta)^4 + 56\cos^3\theta(i\sin\theta)^5 + 28\cos^2\theta(i\sin\theta)^6 \\ &\quad + 8\cos\theta(i\sin\theta)^7 + (i\sin\theta)^8 \\ &= \cos^8\theta + 8i\cos^7\theta\sin\theta - 28\cos^6\theta\sin^2\theta - 56i\cos^5\theta\sin^3\theta \\ &\quad + 70\cos^4\theta\sin^4\theta + 56i\cos^3\theta\sin^5\theta - 28\cos^2\theta\sin^6\theta \\ &\quad - 8i\cos\theta\sin^7\theta + \sin^8\theta\end{aligned}$$

Equating real parts gives

$$\begin{aligned}\cos 8\theta &= \cos^8\theta - 28\cos^6\theta\sin^2\theta + 70\cos^4\theta\sin^4\theta \\ &\quad - 28\cos^2\theta\sin^6\theta + \sin^8\theta\end{aligned}$$

Since $\cos^2\theta = 1 - \sin^2\theta$, let $x = \sin^2\theta$. Then

$$\begin{aligned}\cos 8\theta &= (1-x)^4 - 28(1-x)^3x + 70(1-x)^2x^2 - 28(1-x)x^3 + x^4 \\ &= (1 - 4x + 6x^2 - 4x^3 + x^4) \\ &\quad - 28(x - 3x^2 + 3x^3 - x^4) \\ &\quad + 70(x^2 - 2x^3 + x^4) - 28(x^3 - x^4) + x^4 \\ &= 1 - 32x + 160x^2 - 256x^3 + 128x^4\end{aligned}$$

Substitute back $x = \sin^2\theta$:

$$\cos 8\theta \equiv 128\sin^8\theta - 256\sin^6\theta + 160\sin^4\theta - 32\sin^2\theta + 1$$

Hence

$$a = 128, \quad b = -256, \quad c = 160, \quad d = -32, \quad e = 1$$

(iii) From part (i)(B):

$$\sin\theta = \frac{1}{2i}\left(e^{i\theta} - e^{-i\theta}\right)$$

Therefore

$$\sin^8\theta = \left(\frac{1}{2i}\right)^8\left(e^{i\theta} - e^{-i\theta}\right)^8 = \frac{1}{256}\left(e^{i\theta} - e^{-i\theta}\right)^8$$

Now expand:

$$\sin^8\theta = \frac{1}{256}\Big(e^{8i\theta} - 8e^{6i\theta} + 28e^{4i\theta} - 56e^{2i\theta} + 70 \\ - 56e^{-2i\theta} + 28e^{-4i\theta} - 8e^{-6i\theta} + e^{-8i\theta}\Big)$$

Use

$$e^{ik\theta} + e^{-ik\theta} = 2\cos k\theta$$

so that

$$\begin{aligned}\sin^8\theta &= \frac{1}{256}\Big((e^{8i\theta} + e^{-8i\theta}) - 8(e^{6i\theta} + e^{-6i\theta}) \\ &\qquad + 28(e^{4i\theta} + e^{-4i\theta}) - 56(e^{2i\theta} + e^{-2i\theta}) + 70\Big) \\ &= \frac{1}{256}\Big(2\cos 8\theta - 16\cos 6\theta + 56\cos 4\theta - 112\cos 2\theta + 70\Big) \\ &= \frac{1}{128}\cos 8\theta - \frac{1}{16}\cos 6\theta + \frac{7}{32}\cos 4\theta - \frac{7}{16}\cos 2\theta + \frac{35}{128}\end{aligned}$$

So

$$P = \frac{1}{128}, \quad Q = -\frac{1}{16}, \quad R = \frac{7}{32}, \quad S = -\frac{7}{16}, \quad T = \frac{35}{128}$$

Equivalently

$$\sin^8\theta \equiv \frac{\cos 8\theta - 8\cos 6\theta + 28\cos 4\theta - 56\cos 2\theta + 35}{128}$$

(iv) Using the identity from part (iii) with $\theta = \frac{\pi}{8}$:

$$\sin^8\frac{\pi}{8} = \frac{\cos\pi - 8\cos\frac{3\pi}{4} + 28\cos\frac{\pi}{2} - 56\cos\frac{\pi}{4} + 35}{128}$$

Now

$$\cos\pi = -1, \qquad \cos\frac{3\pi}{4} = -\frac{\sqrt{2}}{2}, \qquad \cos\frac{\pi}{2} = 0, \qquad \cos\frac{\pi}{4} = \frac{\sqrt{2}}{2}$$

so

$$\begin{aligned}\sin^8\frac{\pi}{8} &= \frac{-1 - 8\left(-\frac{\sqrt{2}}{2}\right) + 28(0) - 56\left(\frac{\sqrt{2}}{2}\right) + 35}{128} \\ &= \frac{-1 + 4\sqrt{2} - 28\sqrt{2} + 35}{128} \\ &= \frac{34 - 24\sqrt{2}}{128} \\ &= \frac{17 - 12\sqrt{2}}{64}\end{aligned}$$

Since $0 < \frac{\pi}{8} < \frac{\pi}{2}$, $\sin\frac{\pi}{8} > 0$. Hence

$$\sin\frac{\pi}{8} = \left(\frac{17 - 12\sqrt{2}}{64}\right)^{\frac{1}{8}}$$

17. **(a)** Using de Moivre's theorem, show that

$$\sin^4\theta\cos^2\theta \equiv \frac{1}{32}(\cos 6\theta - 2\cos 4\theta - \cos 2\theta + 2)$$

[5]

(b) By using the substitution $\alpha = \left(\frac{\pi}{2} - \theta\right)$, or otherwise, find a similar identity for $\sin^2\theta\cos^4\theta$. **[3]**

(c) Given that $0 < a < \frac{\pi}{2}$ and

$$\int_0^a \left(\sin^4\theta\cos^2\theta + 2\sin^2\theta\cos^4\theta\right)\,\mathrm{d}\theta = \frac{\pi}{32} - \frac{\sqrt{3}}{64}$$

find the exact value of a. **[5]**

Solution

(a) Let

$$z = \cos\theta + i\sin\theta$$

By de Moivre's theorem:

$$z^n = \cos n\theta + i\sin n\theta, \qquad z^{-n} = \cos n\theta - i\sin n\theta$$

so

$$z^n + z^{-n} = 2\cos n\theta, \qquad z^n - z^{-n} = 2i\sin n\theta$$

In particular, $z + z^{-1} = 2\cos\theta$ and $z - z^{-1} = 2i\sin\theta$, so

$$\begin{aligned}(z - z^{-1})^4(z + z^{-1})^2 &= (2i\sin\theta)^4(2\cos\theta)^2\\ &= 16i^4\sin^4\theta \times 4\cos^2\theta\\ &= 64\sin^4\theta\cos^2\theta\end{aligned}$$

Now expand the left-hand side, using $(z - z^{-1})(z + z^{-1}) = z^2 - z^{-2}$:

$$\begin{aligned}(z - z^{-1})^4(z + z^{-1})^2 &= (z - z^{-1})^2(z^2 - z^{-2})^2\\ &= (z^2 - 2 + z^{-2})(z^4 - 2 + z^{-4})\\ &= z^6 - 2z^2 + z^{-2} - 2z^4 + 4 - 2z^{-4} + z^2 - 2z^{-2} + z^{-6}\\ &= (z^6 + z^{-6}) - 2(z^4 + z^{-4}) - (z^2 + z^{-2}) + 4\\ &= 2\cos 6\theta - 4\cos 4\theta - 2\cos 2\theta + 4\end{aligned}$$

Equating the two expressions:

$$64\sin^4\theta\cos^2\theta = 2\cos 6\theta - 4\cos 4\theta - 2\cos 2\theta + 4$$

so

$$\sin^4\theta\cos^2\theta \equiv \frac{1}{32}(\cos 6\theta - 2\cos 4\theta - \cos 2\theta + 2)$$

Alternative Solution:

Let

$$c = \cos 2\theta$$

Using the double angle formulae:

$$\sin^2\theta = \frac{1-\cos 2\theta}{2} = \frac{1-c}{2}, \qquad \cos^2\theta = \frac{1+\cos 2\theta}{2} = \frac{1+c}{2}$$

So

$$\begin{aligned}\sin^4\theta\cos^2\theta &= \left(\frac{1-c}{2}\right)^2\left(\frac{1+c}{2}\right)\\ &= \frac{1}{8}(1-2c+c^2)(1+c)\\ &= \frac{1}{8}(1+c-2c-2c^2+c^2+c^3)\\ &= \frac{1}{8}(1-c-c^2+c^3)\end{aligned}$$

Now use de Moivre's theorem.

From

$$(\cos 2\theta + i\sin 2\theta)^2 = \cos 4\theta + i\sin 4\theta$$

equating real parts gives

$$\cos 4\theta = 2\cos^2 2\theta - 1 = 2c^2 - 1$$

so

$$c^2 = \frac{1+\cos 4\theta}{2}$$

Also, from

$$(\cos 2\theta + i\sin 2\theta)^3 = \cos 6\theta + i\sin 6\theta$$

equating real parts gives

$$\cos 6\theta = 4\cos^3 2\theta - 3\cos 2\theta = 4c^3 - 3c$$

so

$$c^3 = \frac{\cos 6\theta + 3c}{4}$$

Substituting these into $\frac{1}{8}(1-c-c^2+c^3)$:

$$\begin{aligned}\sin^4\theta\cos^2\theta &= \frac{1}{8}\left(1-c-\frac{1+\cos 4\theta}{2}+\frac{\cos 6\theta+3c}{4}\right)\\ &= \frac{1}{8}\left(\frac{4-4c-2-2\cos 4\theta+\cos 6\theta+3c}{4}\right)\\ &= \frac{1}{32}(2-c-2\cos 4\theta+\cos 6\theta)\end{aligned}$$

Since $c = \cos 2\theta$, this is

$$\sin^4\theta\cos^2\theta \equiv \frac{1}{32}(\cos 6\theta - 2\cos 4\theta - \cos 2\theta + 2)$$

as required.

(b) Let

$$\alpha = \frac{\pi}{2} - \theta$$

Then

$$\sin\theta = \cos\alpha, \qquad \cos\theta = \sin\alpha$$

so

$$\sin^2\theta\cos^4\theta = \cos^2\alpha\sin^4\alpha = \sin^4\alpha\cos^2\alpha$$

Using part (a):

$$\sin^4\alpha\cos^2\alpha = \frac{1}{32}(\cos 6\alpha - 2\cos 4\alpha - \cos 2\alpha + 2)$$

Now

$$2\alpha = \pi - 2\theta, \quad 4\alpha = 2\pi - 4\theta, \quad 6\alpha = 3\pi - 6\theta$$

so

$$\cos 2\alpha = -\cos 2\theta, \qquad \cos 4\alpha = \cos 4\theta, \qquad \cos 6\alpha = -\cos 6\theta$$

Therefore

$$\sin^2\theta\cos^4\theta = \frac{1}{32}(-\cos 6\theta - 2\cos 4\theta + \cos 2\theta + 2)$$

So a similar identity is

$$\sin^2\theta\cos^4\theta \equiv \frac{1}{32}(-\cos 6\theta - 2\cos 4\theta + \cos 2\theta + 2)$$

(c) Let

$$I(a) = \int_0^a \left(\sin^4\theta\cos^2\theta + 2\sin^2\theta\cos^4\theta\right)\mathrm{d}\theta$$

Using parts (a) and (b):

$$\begin{aligned}\sin^4\theta\cos^2\theta + 2\sin^2\theta\cos^4\theta &= \frac{1}{32}(\cos 6\theta - 2\cos 4\theta - \cos 2\theta + 2)\\ &\quad + \frac{2}{32}(-\cos 6\theta - 2\cos 4\theta + \cos 2\theta + 2)\\ &= \frac{1}{32}(-\cos 6\theta - 6\cos 4\theta + \cos 2\theta + 6)\end{aligned}$$

Hence

$$\begin{aligned}I(a) &= \frac{1}{32}\int_0^a (-\cos 6\theta - 6\cos 4\theta + \cos 2\theta + 6)\,\mathrm{d}\theta\\ &= \frac{1}{32}\left[-\frac{1}{6}\sin 6\theta - \frac{3}{2}\sin 4\theta + \frac{1}{2}\sin 2\theta + 6\theta\right]_0^a\\ &= \frac{1}{32}\left(6a + \frac{1}{2}\sin 2a - \frac{3}{2}\sin 4a - \frac{1}{6}\sin 6a\right)\end{aligned}$$

We are given that

$$I(a) = \frac{\pi}{32} - \frac{\sqrt{3}}{64}$$

Try $a = \frac{\pi}{6}$. Then

$$6a = \pi, \qquad \sin 2a = \sin\frac{\pi}{3} = \frac{\sqrt{3}}{2}, \qquad \sin 4a = \sin\frac{2\pi}{3} = \frac{\sqrt{3}}{2}, \qquad \sin 6a = \sin\pi = 0$$

So

$$\begin{aligned} I\Big(\frac{\pi}{6}\Big) &= \frac{1}{32}\left(\pi + \frac{1}{2}\times\frac{\sqrt{3}}{2} - \frac{3}{2}\times\frac{\sqrt{3}}{2} - 0\right) \\ &= \frac{1}{32}\left(\pi - \frac{\sqrt{3}}{2}\right) \\ &= \frac{\pi}{32} - \frac{\sqrt{3}}{64} \end{aligned}$$

So $a = \frac{\pi}{6}$ satisfies the equation.

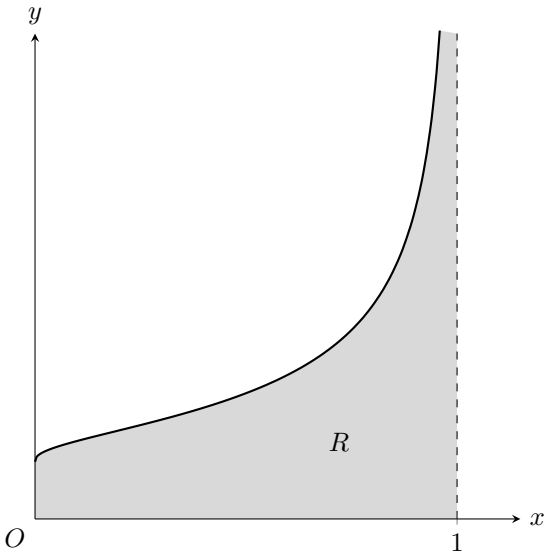
The integrand is $\sin^2\theta\cos^2\theta\big(\sin^2\theta + 2\cos^2\theta\big)$, which is positive for $0 < \theta < \frac{\pi}{2}$, so $I(a)$ increases with a on this interval and the solution is unique.

Therefore

$$a = \frac{\pi}{6}$$

18. **(a)** Use de Moivre's theorem to determine constants A, B and C such that

$$\cos^5\theta \equiv A\cos 5\theta + B\cos 3\theta + C\cos\theta$$ **[5]**



The function f is defined by

$$f(x) = -3\sin\left(5\cos^{-1}\left(x^{\frac{1}{6}}\right)\right) - 25\sin\left(3\cos^{-1}\left(x^{\frac{1}{6}}\right)\right) - 150\sin\left(\cos^{-1}\left(x^{\frac{1}{6}}\right)\right), \qquad 0 \le x < 1$$

(b) Show that

$$f'(x) = \frac{40}{\sqrt{1-x^{\frac{1}{3}}}}$$ **[6]**

The diagram shows the curve with equation

$$y = \frac{1}{\sqrt{1-x^{\frac{1}{3}}}}$$

for $0 \le x < 1$ and the asymptote $x = 1$. The region R is the unbounded region between the curve, the x-axis, the line $x = 0$ and the line $x = 1$.
You are given that the area of R is finite.

(c) Determine the exact area of R. **[3]**

Solution

(a) Let $z = \cos\theta + i\sin\theta$. By de Moivre's theorem:

$$z^n + z^{-n} = (\cos n\theta + i\sin n\theta) + (\cos n\theta - i\sin n\theta) = 2\cos n\theta$$

In particular, $z + z^{-1} = 2\cos\theta$, so

$$(z + z^{-1})^5 = 32\cos^5\theta$$

Expanding by the binomial theorem:

$$\begin{aligned}(z+z^{-1})^5 &= z^5+5z^3+10z+10z^{-1}+5z^{-3}+z^{-5}\\ &= (z^5+z^{-5})+5(z^3+z^{-3})+10(z+z^{-1})\\ &= 2\cos 5\theta+10\cos 3\theta+20\cos\theta\end{aligned}$$

Equating the two expressions and dividing by 32:

$$\cos^5\theta \equiv \frac{1}{16}\cos 5\theta+\frac{5}{16}\cos 3\theta+\frac{5}{8}\cos\theta$$

Hence

$$A=\frac{1}{16},\qquad B=\frac{5}{16},\qquad C=\frac{5}{8}$$

Alternative Solution:

Using de Moivre's theorem:

$$(\cos\theta+i\sin\theta)^5=\cos 5\theta+i\sin 5\theta$$

Expanding the left-hand side:

$$\begin{aligned}(\cos\theta+i\sin\theta)^5 &= \cos^5\theta+5\cos^4\theta(i\sin\theta)+10\cos^3\theta(i\sin\theta)^2\\ &\quad+10\cos^2\theta(i\sin\theta)^3+5\cos\theta(i\sin\theta)^4+(i\sin\theta)^5\\ &= \cos^5\theta+5i\cos^4\theta\sin\theta-10\cos^3\theta\sin^2\theta\\ &\quad-10i\cos^2\theta\sin^3\theta+5\cos\theta\sin^4\theta+i\sin^5\theta\end{aligned}$$

Equating real parts gives

$$\cos 5\theta=\cos^5\theta-10\cos^3\theta\sin^2\theta+5\cos\theta\sin^4\theta$$

Now use $\sin^2\theta=1-\cos^2\theta$:

$$\begin{aligned}\cos 5\theta &= \cos^5\theta-10\cos^3\theta(1-\cos^2\theta)+5\cos\theta(1-\cos^2\theta)^2\\ &= \cos^5\theta-10\cos^3\theta+10\cos^5\theta+5\cos\theta(1-2\cos^2\theta+\cos^4\theta)\\ &= 16\cos^5\theta-20\cos^3\theta+5\cos\theta\end{aligned}$$

Similarly, from

$$(\cos\theta+i\sin\theta)^3=\cos 3\theta+i\sin 3\theta$$

we get

$$\begin{aligned}(\cos\theta+i\sin\theta)^3 &= \cos^3\theta+3\cos^2\theta(i\sin\theta)+3\cos\theta(i\sin\theta)^2+(i\sin\theta)^3\\ &= \cos^3\theta+3i\cos^2\theta\sin\theta-3\cos\theta\sin^2\theta-i\sin^3\theta\end{aligned}$$

so equating real parts gives

$$\begin{aligned}\cos 3\theta &= \cos^3\theta-3\cos\theta\sin^2\theta\\ &= \cos^3\theta-3\cos\theta(1-\cos^2\theta)\\ &= 4\cos^3\theta-3\cos\theta\end{aligned}$$

Now use

$$\cos^5\theta \equiv A\cos 5\theta + B\cos 3\theta + C\cos\theta$$

Substituting the expansions:

$$\begin{aligned}\cos^5\theta &\equiv A(16\cos^5\theta - 20\cos^3\theta + 5\cos\theta) + B(4\cos^3\theta - 3\cos\theta) + C\cos\theta \\ &\equiv 16A\cos^5\theta + (-20A + 4B)\cos^3\theta + (5A - 3B + C)\cos\theta\end{aligned}$$

Equating coefficients of powers of $\cos\theta$:

$$\begin{aligned}16A &= 1 \\ -20A + 4B &= 0 \\ 5A - 3B + C &= 0\end{aligned}$$

So

$$A = \frac{1}{16}$$

$$-20\left(\frac{1}{16}\right) + 4B = 0 \quad \Longrightarrow \quad 4B = \frac{5}{4} \quad \Longrightarrow \quad B = \frac{5}{16}$$

$$5\left(\frac{1}{16}\right) - 3\left(\frac{5}{16}\right) + C = 0 \quad \Longrightarrow \quad -\frac{10}{16} + C = 0 \quad \Longrightarrow \quad C = \frac{10}{16} = \frac{5}{8}$$

Hence

$$A = \frac{1}{16}, \qquad B = \frac{5}{16}, \qquad C = \frac{5}{8}$$

(b) Let

$$\theta = \cos^{-1}\left(x^{1/6}\right)$$

so that

$$\cos\theta = x^{1/6}$$

Then

$$f(x) = -3\sin 5\theta - 25\sin 3\theta - 150\sin\theta$$

Differentiate using the chain rule:

$$\begin{aligned}f'(x) &= -3(5\cos 5\theta)\frac{\mathrm{d}\theta}{\mathrm{d}x} - 25(3\cos 3\theta)\frac{\mathrm{d}\theta}{\mathrm{d}x} - 150(\cos\theta)\frac{\mathrm{d}\theta}{\mathrm{d}x} \\ &= -(15\cos 5\theta + 75\cos 3\theta + 150\cos\theta)\frac{\mathrm{d}\theta}{\mathrm{d}x}\end{aligned}$$

Now

$$\theta = \cos^{-1}(x^{1/6})$$

and

$$\frac{\mathrm{d}}{\mathrm{d}x}\left(\cos^{-1}u\right) = -\frac{1}{\sqrt{1-u^2}}\frac{\mathrm{d}u}{\mathrm{d}x}$$

Here $u = x^{1/6}$, so

$$\frac{\mathrm{d}u}{\mathrm{d}x} = \frac{1}{6}x^{-5/6}$$

and therefore

$$\begin{aligned}\frac{\mathrm{d}\theta}{\mathrm{d}x} &= -\frac{1}{\sqrt{1-(x^{1/6})^2}} \times \frac{1}{6}x^{-5/6} \\ &= -\frac{x^{-5/6}}{6\sqrt{1-x^{1/3}}}\end{aligned}$$

Substitute this into the expression for $f'(x)$:

$$\begin{aligned}f'(x) &= -(15\cos 5\theta + 75\cos 3\theta + 150\cos\theta)\left(-\frac{x^{-5/6}}{6\sqrt{1-x^{1/3}}}\right) \\ &= \frac{x^{-5/6}}{6\sqrt{1-x^{1/3}}}(15\cos 5\theta + 75\cos 3\theta + 150\cos\theta) \\ &= \frac{5}{2} \times \frac{x^{-5/6}}{\sqrt{1-x^{1/3}}}\left(\cos 5\theta + 5\cos 3\theta + 10\cos\theta\right)\end{aligned}$$

From part (a):

$$\cos^5\theta = \frac{1}{16}\cos 5\theta + \frac{5}{16}\cos 3\theta + \frac{5}{8}\cos\theta$$

Multiply by 16:

$$16\cos^5\theta = \cos 5\theta + 5\cos 3\theta + 10\cos\theta$$

So

$$f'(x) = \frac{5}{2} \times \frac{x^{-5/6}}{\sqrt{1-x^{1/3}}} \times 16\cos^5\theta$$

But $\cos\theta = x^{1/6}$, so

$$\cos^5\theta = \left(x^{1/6}\right)^5 = x^{5/6}$$

Hence

$$\begin{aligned}f'(x) &= \frac{5}{2} \times \frac{x^{-5/6}}{\sqrt{1-x^{1/3}}} \times 16x^{5/6} \\ &= \frac{40}{\sqrt{1-x^{1/3}}}\end{aligned}$$

(c) The area of R is

$$\int_0^1 \frac{1}{\sqrt{1-x^{1/3}}}\,\mathrm{d}x$$

Since the curve has asymptote $x = 1$, this is an improper integral:

$$\text{Area of } R = \lim_{b\to 1^-}\int_0^b \frac{1}{\sqrt{1-x^{1/3}}}\,\mathrm{d}x$$

From part (b):

$$\frac{1}{\sqrt{1-x^{1/3}}} = \frac{1}{40} f'(x)$$

So

$$\begin{aligned}
\text{Area of } R &= \frac{1}{40} \lim_{b \to 1^-} \int_0^b f'(x)\,\mathrm{d}x \\
&= \frac{1}{40} \lim_{b \to 1^-} \big[f(x)\big]_0^b \\
&= \frac{1}{40} \left(\lim_{b \to 1^-} f(b) - f(0) \right)
\end{aligned}$$

First find $\lim_{b \to 1^-} f(b)$.

As $b \to 1^-$:

$$b^{1/6} \to 1 \qquad \text{so} \qquad \cos^{-1}(b^{1/6}) \to 0$$

and therefore

$$f(b) \to -3\sin 0 - 25\sin 0 - 150\sin 0 = 0$$

Now find $f(0)$:

$$\cos^{-1}(0^{1/6}) = \cos^{-1}(0) = \frac{\pi}{2}$$

so

$$\begin{aligned}
f(0) &= -3\sin\left(5 \times \frac{\pi}{2}\right) - 25\sin\left(3 \times \frac{\pi}{2}\right) - 150\sin\left(\frac{\pi}{2}\right) \\
&= -3(1) - 25(-1) - 150(1) \\
&= -3 + 25 - 150 \\
&= -128
\end{aligned}$$

Hence

$$\begin{aligned}
\text{Area of } R &= \frac{1}{40}\big(0 - (-128)\big) \\
&= \frac{128}{40} \\
&= \frac{16}{5}
\end{aligned}$$